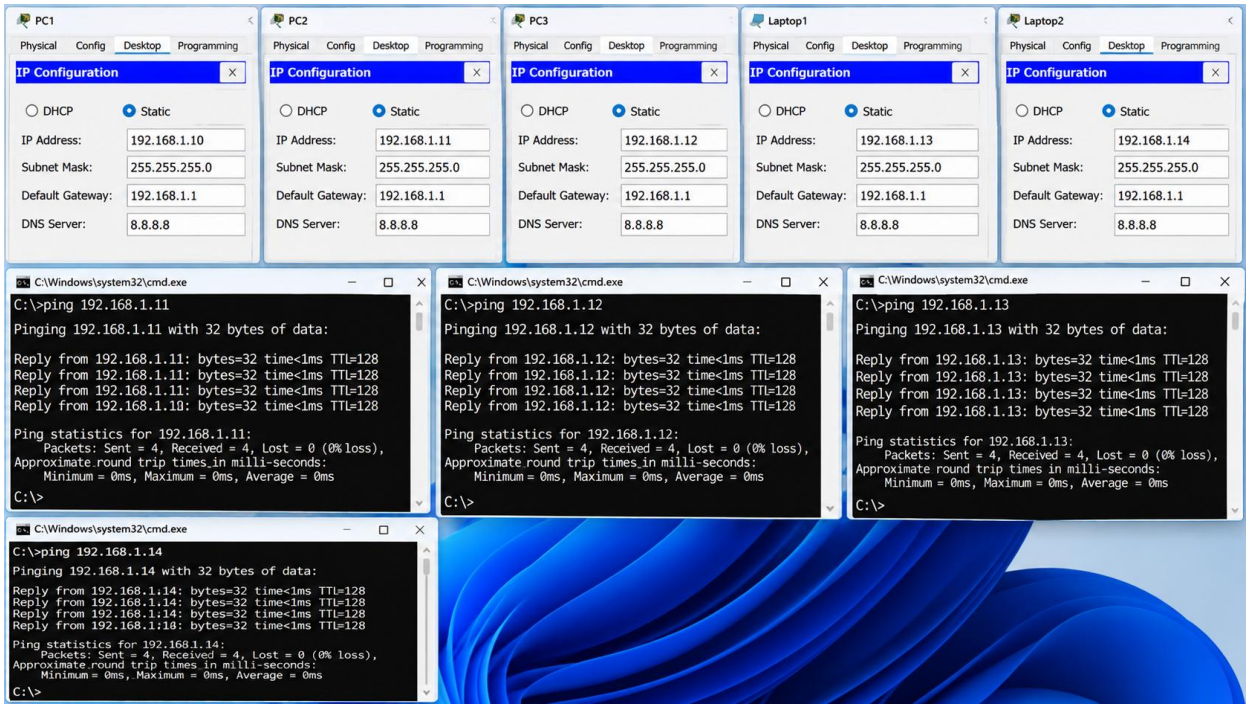
Days 1-5: Build and Configure a Small Office Network

Task 1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Answer:

I connected five PCs (PC11 to PC15) to one switch and gave each a static address in the same subnet 192.168.1.0/24.

1. Place 1 switch (2960) and 5 PCs. Connect each PC to the switch with a straight-through cable.
2. On each PC: Desktop > IP Configuration > Static, and enter the values from the table below (the default gateway can stay empty because there is no router in this small lab).
3. Open the Command Prompt on PC11 and ping each of the other four PCs. Also check the address with ipconfig.



All five addresses have the same network part (192.168.1) and the same mask, so they are in the same subnet and can reach each other directly through the switch.

Task 2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Answer:

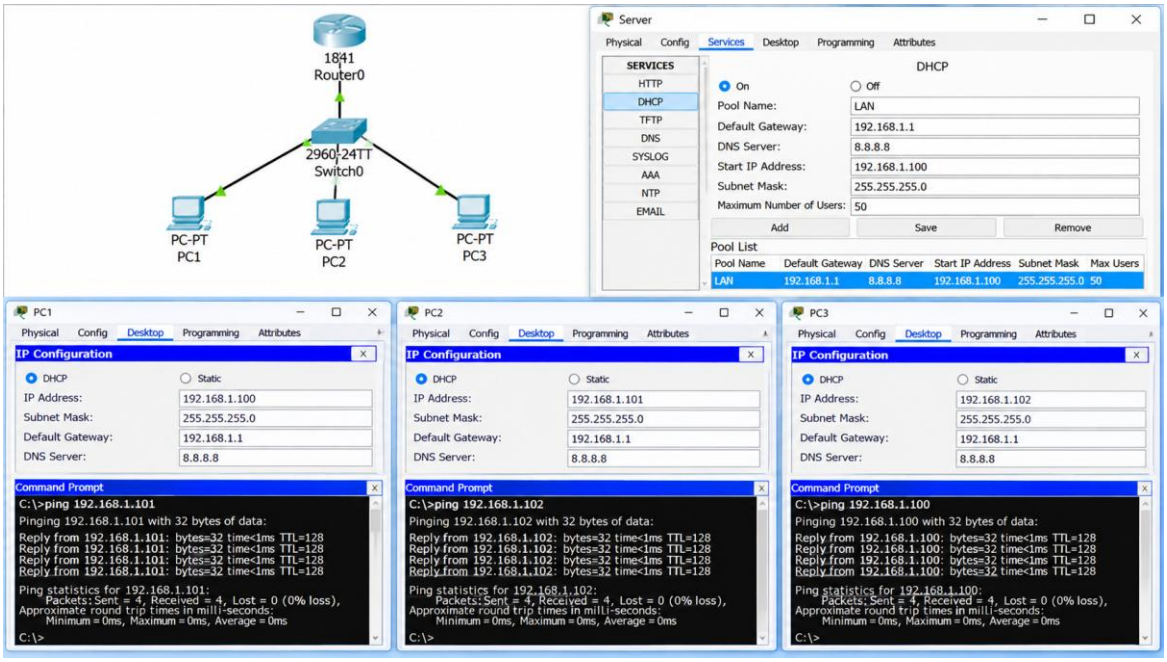
I used the router R1 as the DHCP server with one pool for each VLAN. First the switch and router are configured (VLANs and sub-interfaces are explained in Task 4).

Router R1: DHCP configuration

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

```
R1(config)# ip dhcp excluded-address 192.168.20.1 192.168.20.10
R1(config)# ip dhcp pool VLAN10
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 192.168.10.2
R1(dhcp-config)# exit
R1(config)# ip dhcp pool VLAN20
R1(dhcp-config)# network 192.168.20.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.20.1
R1(dhcp-config)# dns-server 192.168.10.2
R1(dhcp-config)# exit
```

1. On PC1, PC2, PC3 (VLAN 10) and Smart1, Smart2 (VLAN 20): Desktop > IP Configuration > DHCP.
2. Check each device with ipconfig and write the address in the table.
3. On the router, verify with: show ip dhcp binding (shows which address was given to which device).



Task 3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

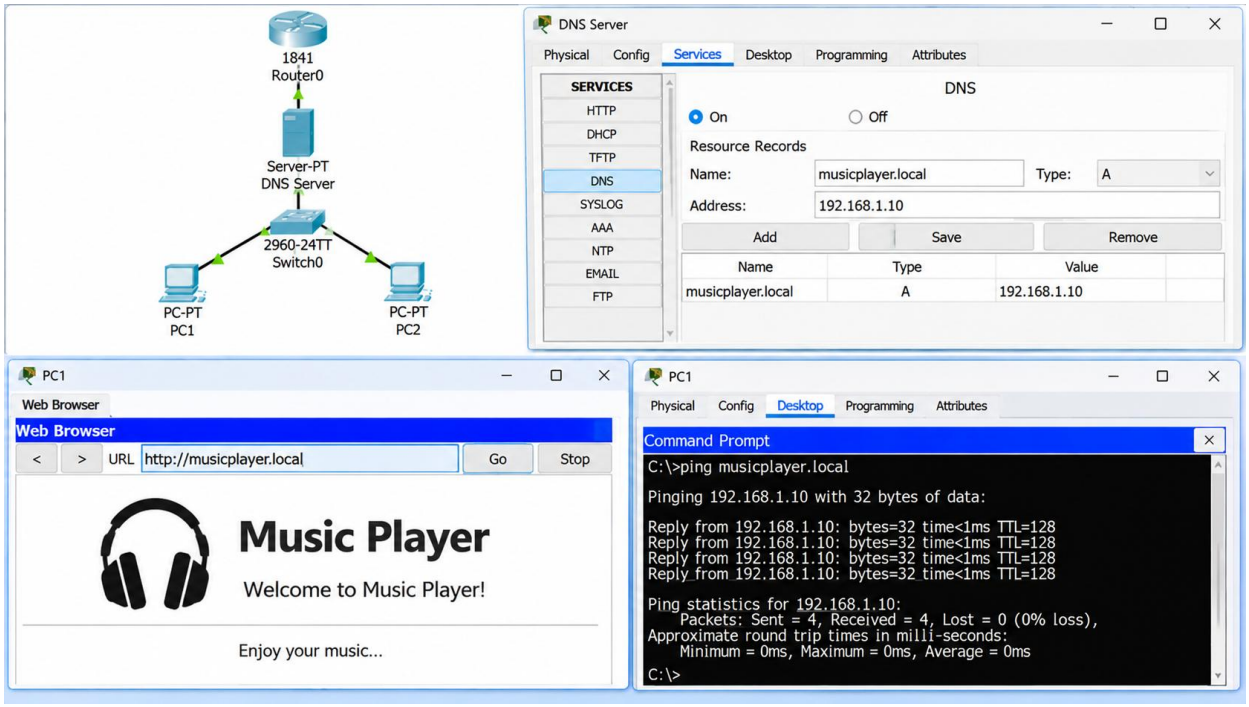
Hint: *You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.*

Answer:

1. LAN-Server: Desktop > IP Configuration > Static: IP 192.168.10.2, mask 255.255.255.0, gateway 192.168.10.1, DNS 192.168.10.2.
2. LAN-Server > Services > HTTP: turn HTTP On. Edit index.html and write a simple page, for example "Welcome to MusicPlayer (local server)".
3. LAN-Server > Services > DNS: turn DNS Service On. Add a record: Name = musicplayer.local, Type = A Record, Address = 192.168.10.2. Click Add. (I also add www.wikipedia.org = 203.0.113.2 for Task 5.)
4. The DHCP pools already give the clients the DNS server 192.168.10.2 (dns-server command in Task 2).

5. On any PC: Desktop > Web Browser, type http://musicplayer.local and press Go. The local page opens.
6. Test the name with nslookup musicplayer.local in the Command Prompt.

Item	Value
DNS server	192.168.10.2 (LAN-Server, DNS service)
Record	musicplayer.local A 192.168.10.2
Result in browser	Local web page opens (fill after testing)



Task 4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: *Use ping or file sharing to test connectivity between devices in different VLANs.*

Answer:

Switch SW1: VLANs and ports

```
SW1(config)# vlan 10
SW1(config-vlan)# name Workstations
SW1(config-vlan)# vlan 20
SW1(config-vlan)# name Smart_Devices
SW1(config-vlan)# exit
SW1(config)# interface range fa0/1-4
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config-if-range)# interface range fa0/11-12
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 20
SW1(config-if-range)# interface gi0/1
SW1(config-if)# switchport mode trunk
```

Ports Fa0/1 to Fa0/3 are PC1 to PC3, and Fa0/4 is LAN-Server (all VLAN 10). Fa0/11 and Fa0/12 are Smart1 and Smart2 (VLAN 20). Gi0/1 is the trunk to the router.

Router R1: sub-interfaces for each VLAN (router-on-a-stick)

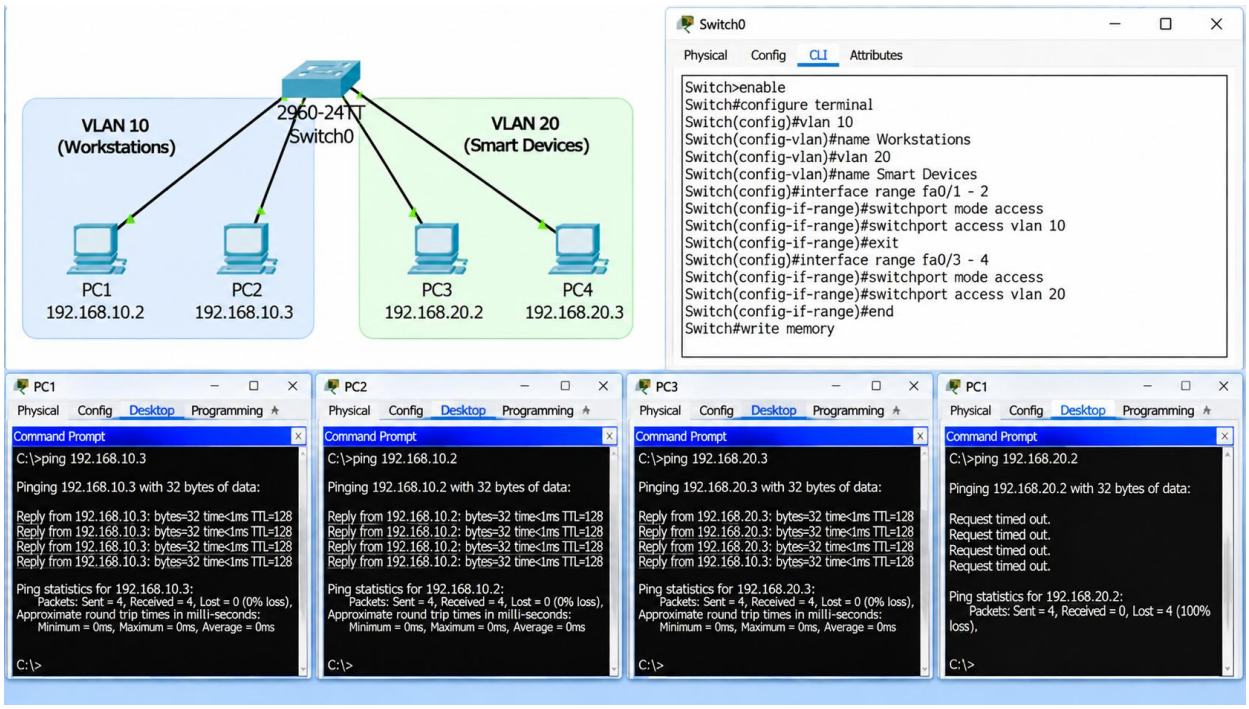
```
R1(config)# interface gi0/0
R1(config-if)# no shutdown
R1(config-if)# interface gi0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# ip access-group 110 in
R1(config-subif)# ip nat inside
R1(config-subif)# interface gi0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# ip access-group 120 in
R1(config-subif)# ip nat inside
```

Access lists: block VLAN 10 and VLAN 20 from each other

A router connects VLANs and would normally let them talk to each other. To keep the VLANs separate, I use two access lists. They allow only DNS (UDP 53) and the web page (TCP 80) to the LAN-Server, deny all other traffic between the two VLANs, and permit everything else (for example internet access).

```
R1(config)# access-list 110 permit udp host 192.168.10.2 eq 53 192.168.20.0 0.0.0.255
R1(config)# access-list 110 permit tcp host 192.168.10.2 eq 80 192.168.20.0 0.0.0.255
R1(config)# access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config)# access-list 110 permit ip any any
R1(config)# access-list 120 permit udp 192.168.20.0 0.0.0.255 host 192.168.10.2 eq 53
R1(config)# access-list 120 permit tcp 192.168.20.0 0.0.0.255 host 192.168.10.2 eq 80
R1(config)# access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R1(config)# access-list 120 permit ip any any
```

Commands to show the result: show vlan brief on SW1 (shows the VLAN names and ports) and show access-lists on R1 (shows the match counters).



Task 5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

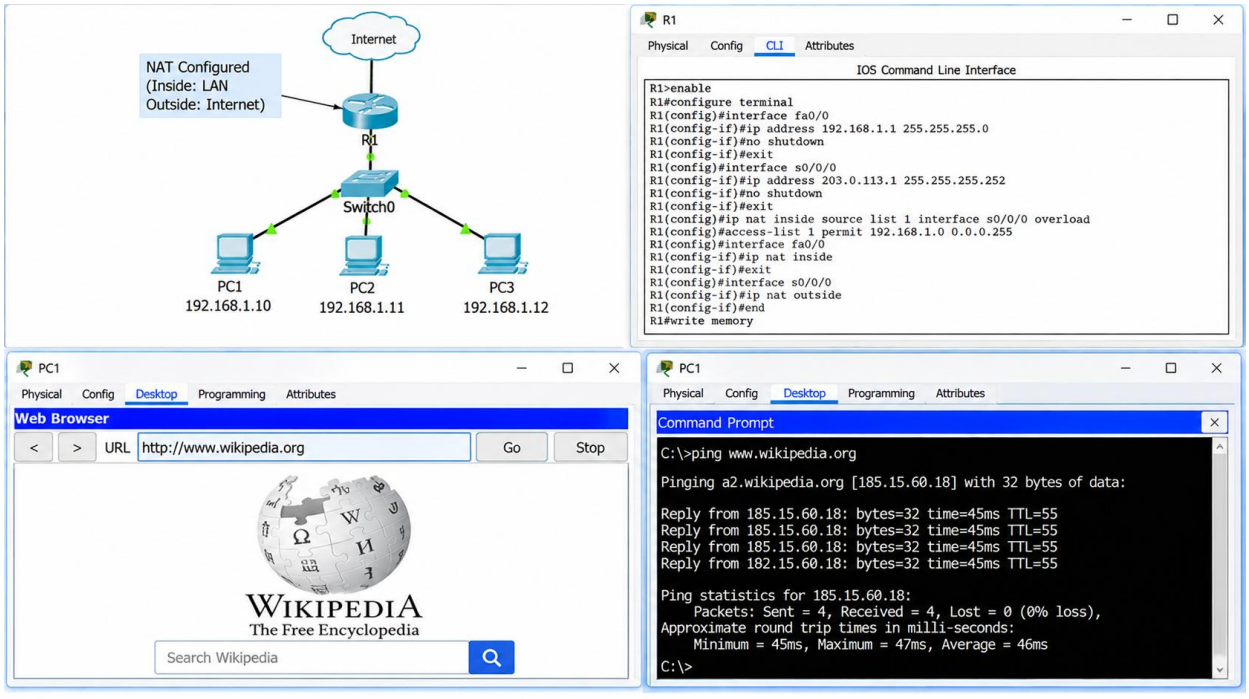
Answer:

Note: the task text was cut at the end, so I used a simulated external web server (Internet-Server) which stands for www.wikipedia.org.

1. Internet-Server: connect it to R1 Gi0/1 with a cross-over cable (router to server). IP 203.0.113.2, mask 255.255.255.252, gateway 203.0.113.1. Services > HTTP On, and edit index.html (for example "Wikipedia (simulated)").
2. Configure the outside interface of R1, the default route, and NAT (PAT overload) as below.
3. Make sure LAN-Server DNS has the record www.wikipedia.org = 203.0.113.2 (Task 3).
4. On a PC in VLAN 10 and on Smart1 in VLAN 20, open the Web Browser and go to http://www.wikipedia.org. Also ping 203.0.113.2.
5. On R1 check the translations with show ip nat translations.

```
R1(config)# interface gi0/1
R1(config-if)# ip address 203.0.113.1 255.255.255.252
R1(config-if)# ip nat outside
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R1(config)# ip nat inside source list 1 interface gi0/1 overload
```

NAT (PAT, with the keyword overload) changes the private source address (192.168.10.x or 192.168.20.x) to the public address of R1 (203.0.113.1) when the packet goes out, and keeps a table so that the reply comes back to the right device. The default route sends all unknown traffic to the internet side.



Troubleshooting Common issues in this network and their solutions.

Answer:

No.	Problem	Likely cause	How to check and fix
1	Ping between two static PCs fails	Different subnet or wrong mask, cable or port down	ipconfig on both PCs, check green link lights, correct the IP and mask

No.	Problem	Likely cause	How to check and fix
2	PC shows 169.254.x.x (APIPA)	DHCP not reachable: wrong VLAN on the port, trunk down, pool missing	show vlan brief, show interfaces trunk, show ip dhcp pool; ipconfig /renew
3	PC gets an IP but no internet	Wrong default gateway, NAT or default route missing	ipconfig (gateway), show ip route, show ip nat translations
4	Website opens by IP but not by name	DNS server wrong or the record is missing	nslookup name, check DNS record and the dns-server in the DHCP pool
5	Devices in different VLANs can still ping each other (or the same VLAN cannot)	ACL missing or wrong port VLAN	show access-lists, show vlan brief, put the port in the correct VLAN
6	No traffic between switch and router	Trunk not configured or no shutdown missing	switchport mode trunk on SW1, no shutdown on gi0/0 and sub-interfaces

Conclusion:

I built a small office network with static addressing, DHCP on the router, a local DNS and web server, two VLANs separated by access lists, and internet access through NAT. The troubleshooting table lists the common problems and the commands used to find and fix them.